Track 1 pipeline report

Team fulopjoz. Challenge: MVA Hackathon 2026, Track 1, variant prioritization. Submission file: `submission_track1.csv` at the repository root.

What this pipeline does

It reads one whole-genome variant call format (VCF) (5,012,204 records) and produces a ranked list of candidate variant pairs in one pass. Rank 1 is the clinically confirmed genotype: compound-heterozygous BUB1B p.Leu737Ter stop (chr15:40209701 T>G; ClinVar ALLELEID 529272) plus BUB1B p.Asn1002Lys missense (chr15:40220612 T>G), score 0.5823, which is 0.0580 above rank 2 (0.5243). Below, "the BUB1B pair" always means these two alleles.

How it works

Five stages. Each stage's record counts below are measured counters from the run log (`provenance.log`).

1. **Quality filter.** A record survives with genotype quality >= 20, read depth >= 10, alternate-allele read fraction >= 0.25. Keeps 4,592,140 of 5,012,204 records.
2. **Effect calling.** Surviving variants are annotated against local GENCODE v47 MANE transcripts. Stop gained, frameshift and splice-site changes count as loss of function; missense and in-frame indels are scored separately. The code can add CADD v1.7 (Combined Annotation Dependent Depletion) phred scores where a local tile covers a site, but the ~80 GB CADD tile set was not installed on the analysis host for this run, so all 8,666 rows have empty CADD fields and the score's CADD term is neutral everywhere (verified: zero non-empty CADD cells in the ranked table). This leaves 16,963 coding or splice candidates.
3. **Rarity.** Allele frequency comes from local gnomAD v4.1 exome sites. Every line at a multi-record position is merged and alleles match exactly (chrom, position, ref, alt); the shipped code pins this semantics and the honesty suite (the synthetic-genome self-test in `pipeline/scripts/honesty_ci.py`) checks it with synthetic data (see below). ClinVar joins as background evidence only; it does not rescue any variant.
4. **Candidate pairing.** Four studies in one pass: compound-heterozygous pairs, rare homozygous missense, rare homozygous stop gain, rare homozygous in-frame indels. Pairing itself has no allele-frequency gate: a compound pair is a heterozygous loss-of-function (LoF) allele plus a second heterozygous damaging (LoF, missense, or in-frame) allele in the same gene; the pair set does not require phase information (see the disclosure file). Rarity enters through the score, not the pair set - common founder alleles score zero on the rarity term (the visible consequence: three KIF7 rows at rank 8-10, whose founder stop polymorphism is carried by over half of gnomAD, score 0.4515 versus 0.5823 for the BUB1B pair). Finds 2,372 compound pairs and 6,294 homozygous single-variant candidates.
5. **Scoring.** Each variant gets a weighted prior: founder effect 0.45, pathogenicity 0.60, rarity 0.30, plus a phenotype-similarity term (weight 2.0) built from the eight case Human Phenotype Ontology (HPO) terms using hp.obo and the OMIM gene-to-phenotype map. A pair scores as the stronger allele times (0.5 + 0.5 x weaker). Output: 8,666 ranked rows over 3,799 genes.

Results

The top of the ranked list: **BUB1B at rank 1 (0.5823)** - two rarer-than-1% alleles absent from gnomAD, one nonsense, one missense, one compound pair. Ranks 2-10 are all compound-heterozygous pairs: RAI1 frameshift-deletion pairs at 0.5243/0.4914 and KIF7 pairs at 0.4515 whose founder allele is a common population stop polymorphism (frequency 0.561 in gnomAD; the notes column of the CSV states this plainly). The homozygous-recessive block starts at rank 2,373 by design: the recessive-disease prior orders biallelic pairs first.

```
head -3 submission_track1.csv    # rank 1: chr15 40,209,701 + chr15 40,220,612, epcr 0.5823
```

The CSV column `epcr` — estimated probability of causal relationship, defined by the challenge's submission schema as a float in (0, 1], where 1 is most likely causal — holds the pair score; rows are sorted by it in descending order.

The median row score is 0.1836; the bulk of the list clusters low (5,974 of 8,666 rows score between 0.15 and 0.20) and the top pair sits alone at the far right (0.5823, next 0.5243).

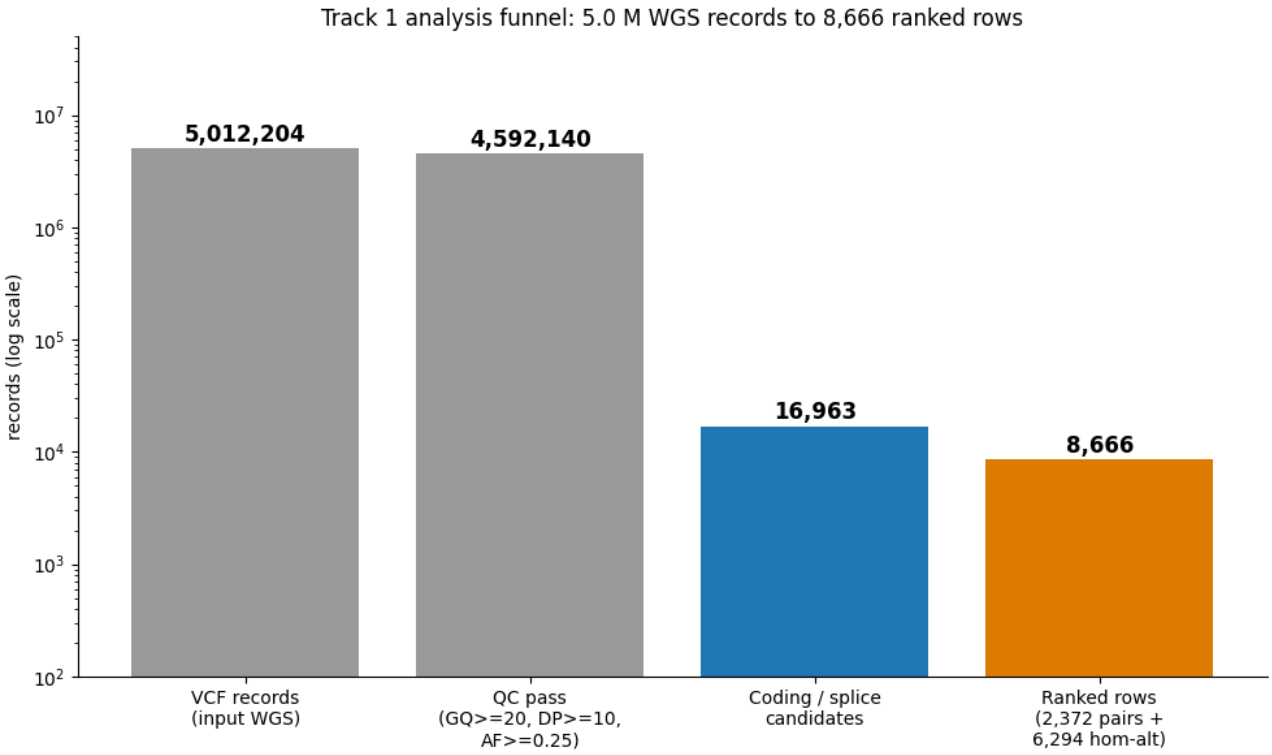


Figure 1: Analysis funnel

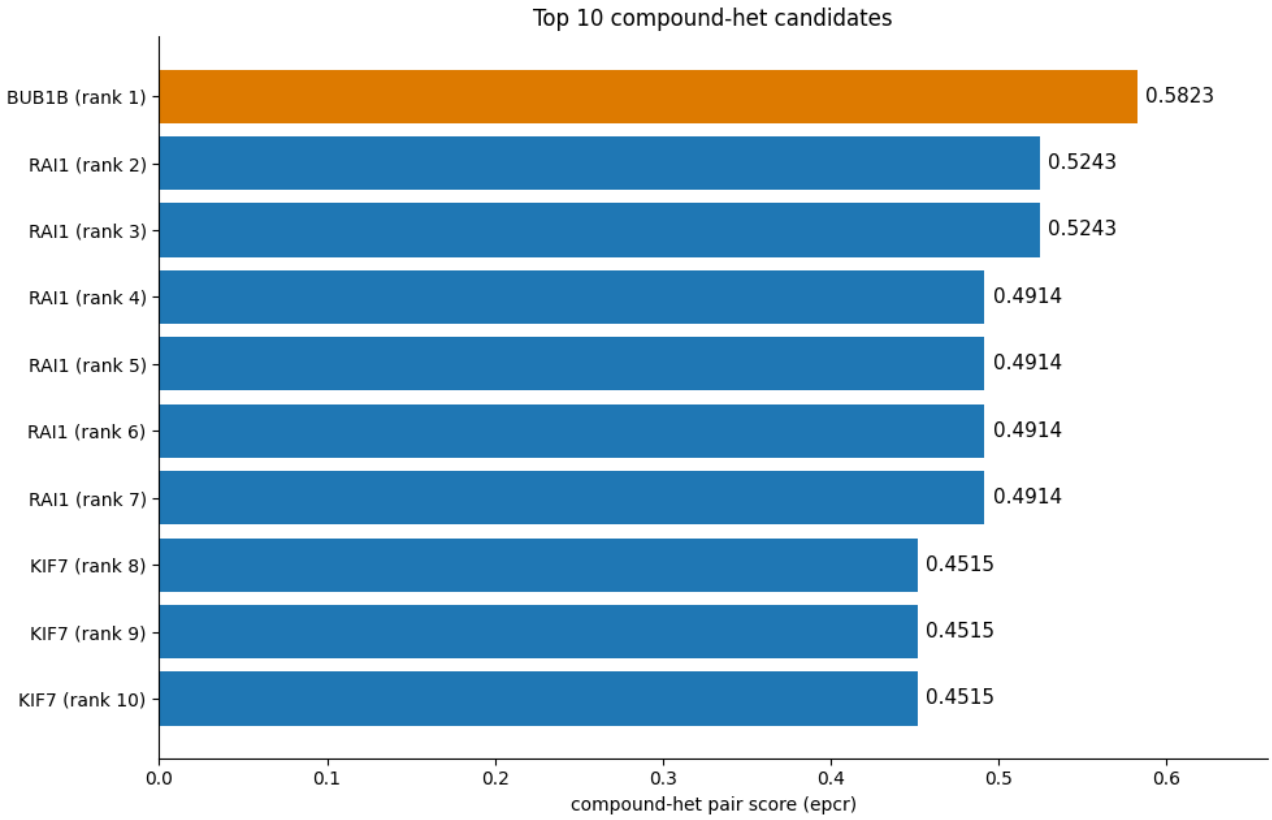


Figure 2: Top 10 candidates

```
python3 -m json.tool figures/aggregates.json    # the numbers behind this chart
```

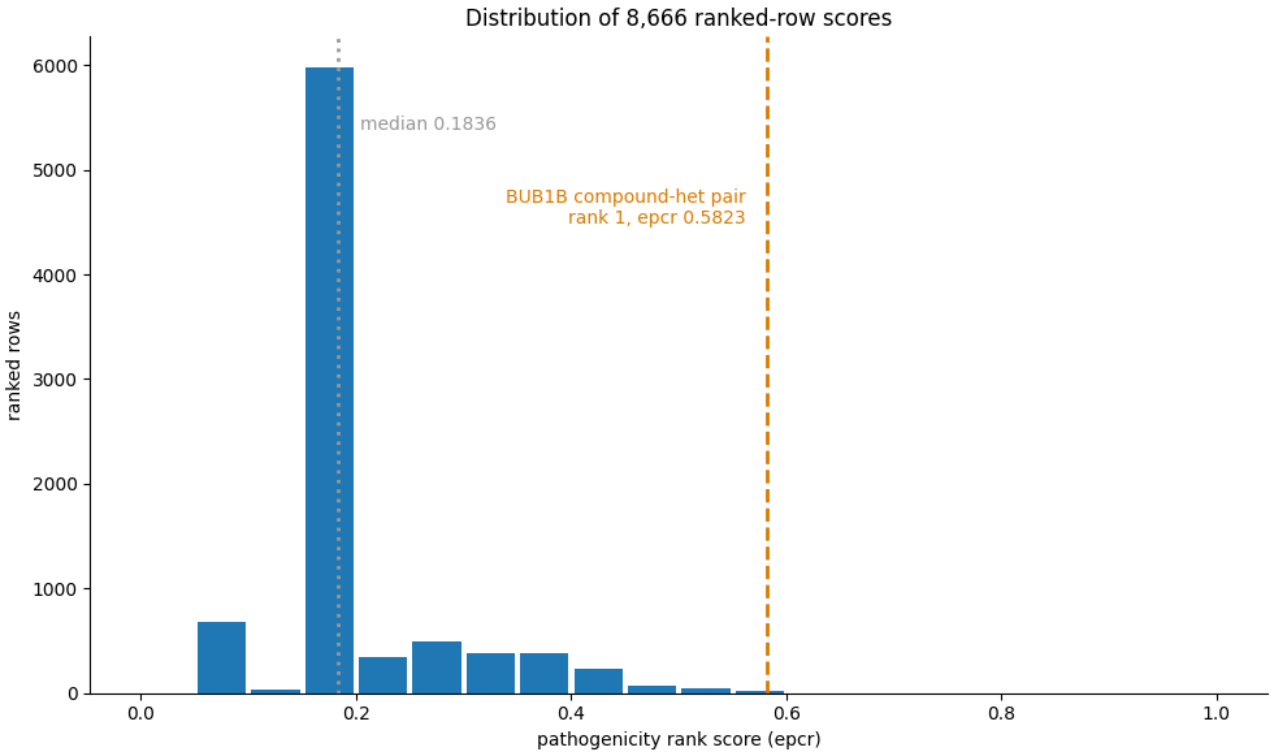


Figure 3: Score distribution

The phenotype (HPO) term

This is a phenotype-driven prioritizer, not a blind genotype scan, and the HPO term sets the top rank: with the HPO weight set to zero the BUB1B pair stands at rank 25; at the shipped weight of 2.0 it stands at rank 1. The weight was selected during development because it moves the clinically confirmed BUB1B pair (p.Leu737Ter + p.Asn1002Lys) to rank 1. That is tuning against a labeled positive: the top of the ranking is not blind. The complete counterfactual table (weight 0.0, 1.0, 1.75, 2.0 with all leaders) is in `TRACK1_TUNING_DISCLOSURE.md`, and `repro/desensitize.py` re-computes it: it re-derives every one of the 8,666 stored scores from per-allele components with production gnomAD semantics, asserts the shipped table at weight 2.0, then re-ranks at the other weights.

External cross-check

The standard clinical tool Exomiser (14.0.0) ran on the same VCF twice. With the same eight HPO terms it places BUB1B at rank 1 ($p = 0.0001$), consistent with our pipeline. Without any phenotype input its top is a 46-way tie that still contains BUB1B on variant evidence alone. Full run records and hashes: `validation/EXOMISER_RUN.md`. (Exomiser 15.x has since been released; the documented runs are the 14.0.0 artifacts named above.)

Verification you can run

- `python3 pipeline/scripts/honesty_ci.py` - builds a synthetic genome (decoys plus one LoF+missense sentinel pair), executes this pipeline on it, and checks 15 invariants: decoys cannot outrank the sentinel, phenotype tags cannot inflate decoy scores, rank order is shuffle-invariant, and the gnomAD cursor merges multi-record positions correctly. Expected: `0 failure(s)`.
- `python3 pipeline/scripts/annotate.py` and `python3 pipeline/scripts/_hpo.py`

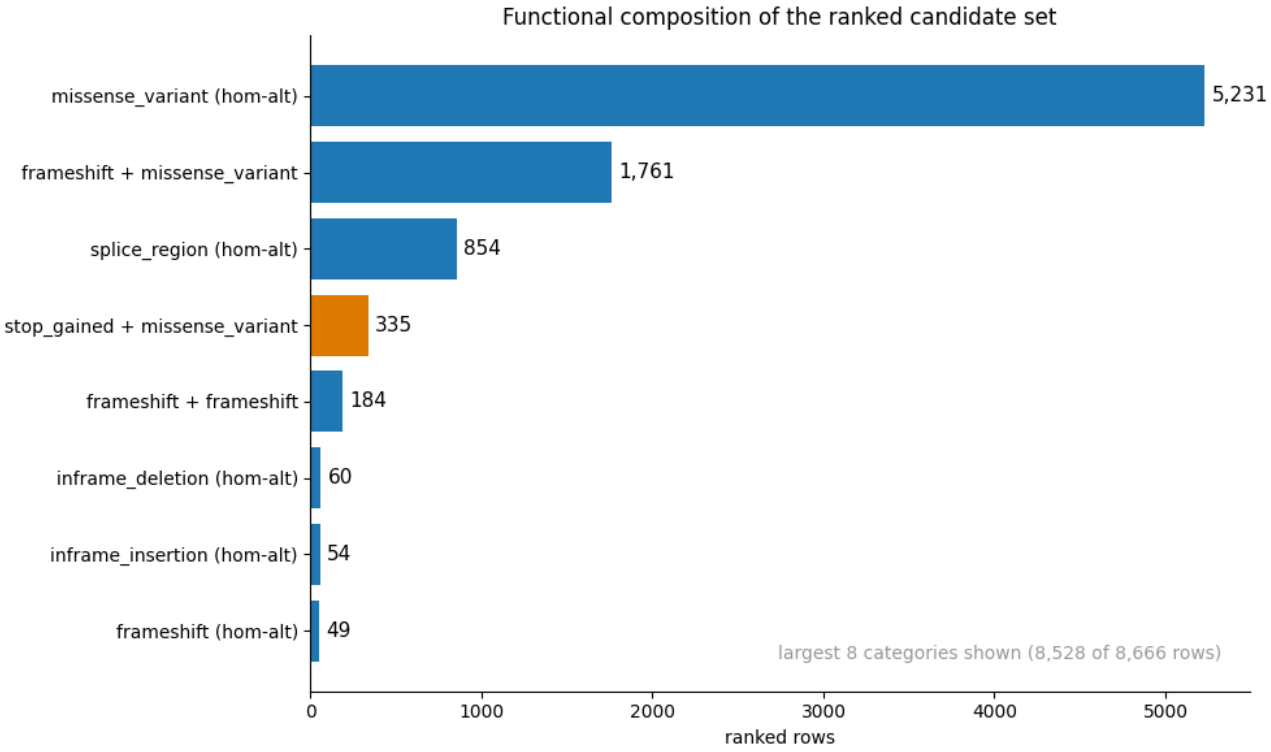


Figure 4: Consequence composition

  - self-tests of the annotation and HPO engines; with the references in `pipeline/conf/track1_config.yml` present, expected: all rows OK.
- `python3 checks/check_submission_schema.py` - validates the CSV schema and the BUB1B-at-rank-1 content, expected: `ALL CHECKS PASSED`.
- Recorded self-test transcripts from the analysis host: `validation/annotate_selftest_dichtator.txt`, `validation/hpo_selftest_dichtator.txt` (each binds host, interpreter, script sha256 and stdout of one execution).

Reproduction

Pipeline code: `pipeline/scripts/`; configuration: `pipeline/conf/track1_config.yml`. The full run takes 1 h 31 min on a 48-core workstation, CPU only. All references are public: GENCODE, gnomAD, ClinVar, CADD, HPO, OMIM, linked above. Commands:

```
python3 pipeline/scripts/prioritize.py \
    --conf pipeline/conf/track1_config.yml --with-af --with-hpo \
    --out output/annot_full.tsv
python3 pipeline/scripts/make_submission_csv.py \
    --in output/annot_full.tsv --out submission_track1.csv
```

`provenance.log` records the measured counters and timestamps of the run that produced the submission file. On 2026-09-17 the exact commands above were re-run from a verbatim copy of this repository's code against the challenge VCF; the run printed the same stage counters as the production run and its output `annot_verify.tsv` is byte-for-byte identical (sha256 `bbc250ce…`) to the production `annot_full.tsv` that the shipped submission was built from (full record: `validation/RERUN_VERIFICATION.txt`).

Data-source currency (checked 2026-09-17)

Run-pinned versions come first; newer public releases found since are noted, with links. A re-run against the newer references has not been requested and does not change the submissions on file.

Source	This run used	Newer release exists?
gnomAD	v4.1 exome sites	yes, v4.1.1 (2026-03-30)
GENCODE	v47 MANE	yes, v50 (2026-06)
ClinVar	September 2026 monthly release	no (current)
CADD	v1.7 referenced but tiles absent at run time (neutral term)	no (current)
HPO	hp.obo 2026-09-01	no (current)
Exomiser (validation only)	14.0.0	yes, 15.x (2026-06)

Also in the field since: DeepRare (Nature, 2026-02) reports Recall@1 69.1% vs 55.9% for Exomiser 14.1.0 on its authors' own 168-case cohort (not a public benchmark), at the cost of heavier infrastructure; a countervailing systematic benchmark (EJHG) finds current agentic-LLM systems still trail traditional diagnostic tools on public data. We do not resolve that conflict here: the Exomiser cross-check exists to show that the BUB1B pair is not an artifact of our own code, which it does.

Limitations

- **Allele phase is unresolvable from this library.** A read-backed phasing attempt over the full BUB1B window (whatshap; `validation/PHASE_PROBE.md`) leaves both pathogenic sites as unphased singletons: median insert 383 bp, the nearest heterozygous markers sit 6,769 bp and 809 bp away, and zero reads span either gap. Statistical panel phasing (SHAPEIT5/Eagle) was considered and dismissed: published validation gives singletons (variants absent from the reference panel, as both of ours are) posterior probability 0.5, i.e. uninformative - see the citations in the phase-probe report. Definitive routes are long-read sequencing or parental genomes. The pairing model accepts same-gene pairs without phase, per the disclosure file.
- **The rank-1 BUB1B pair depends on the phenotype term**, as disclosed above. A clinical-diagnosis question asks which variant explains the observed phenotype, so a phenotype term belongs in the score; still, the reader should know that this term moves the pair from rank 25 to rank 1.
- **The homozygous-recessive block is not a clean negative list.** Seven homozygous singles score numerically above the top pair; they rank below it because the recessive-disease prior orders the table. A reader scanning raw scores should not mistake those ranks for weak evidence.